POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2021
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Signal Analysis and Signal Processing. What are the advantages of Digital signal processing over Analog signal processing? 8
   b) The output of an LTI system is y (n) = {2, 5, 1, -10, -10, -3, 6, 9}to an impulse response h(n) = {1, 2, 0, -3}.Find the input x(n) using convolution sum. The underline represents the element at n = 0.
2. a) Find the convolution between two signals
   $x[n] = u[n]$ and $h[n] = a^n u[n],\ 0 < a < 1$
   b) Determine Z–transform along with ROC of
   $$x(n) = \left[\left(\frac{1}{2}\right)^n + \left(\frac{3}{4}\right)^n\right] u(n-5)$$
3. a) Find the 8-point DITFFT of $x(n) = sin\frac{3\pi n}{4}$, $for\ 0 \le n \le 7$ 8
   b) What is Zero Padding ? Find the circular Convolution of $x_1[n] = \{1,2\}$ & $x_2[n] = \{3,2,1\}$ 7
4. a) Obtain the parallel form realization of following IIR filter 7
   $$H(z) = \frac{3(2z^2 + 5z + 4)}{(2z + 1)(z + 2)}$$
   b) Compute Lattice coefficients and draw lattice structure for given IIR system. Also check stability of the system.
   $H(z) = 1 / (1 - 0.525z^{-1} + 0.6125z^{-2} + 0.3z^{-3})$
5. a) Design a lowpass filter which will have -3 dB cut off at 30π rad/sec and an attenuation of 50 dB at 45π rad/sec. The filter is required to have a linear phase and the system uses a sampling rate of 100 samples/sec. 8
   b) Design a linear FIR filter using Kaiser window to meet the following specifications: 7

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$$0.99 \leq |H(e^{j\omega})| \leq 1.01, \quad for\ 0 \leq |\omega| \leq 0.19\pi$$
$$\leq |H(e^{j\omega})| \leq 0.01, \quad for\ 0.21\pi \leq |\omega| \leq \pi$$

6. a) Determine H(z) using impulse invariance method for the following analog filter's transfer function : 7

$$H_a(s) = \frac{1}{(s+0.5)+(s^2+0.5s+2)}$$

b) Design a digital lowpass Butterworth filter using Bilinear Transformation method to meet the following specifications: 8
Passband attenuation = 1 dB
Stopband attenuation = 16 dB
Passband edge = 120 Hz
Stopband edge = 170 Hz
Sampling frequency = 256 Hz

7. Write short notes on: (Any two) 2×5
   a) Region of Convergence
   b) Symmetric and Anti-symmetric filter
   c) Kaiser window

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THE LEARNING HUB

POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2022
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Consider the analog signal $x_a(t) = 3\cos(100\pi t)$ 8

   i) Determine the minimum sampling rate required to avoid aliasing.

   ii) What is the Nyquist rate for the signal $X_a(t)$?

   iii) If the signal is sampled at the rate F_s = 200Hz. What is the discrete-time signal obtain after sampling?

   iv) What is the frequency $0 < F < F_s/2$ of a sinusoid that yields samples identical to those obtained for F_s=200Hz

   b) Define time-invariance and causality of a discrete time system. Examine whether following systems are stable or not: 7

   i) $y[n] = x[n] + 1$

   ii) $y[n] - y[n-1] = x[n] + x[n-1]$

2. a) Find the convolution between two signals $x[n] = a^n, for\ 0 \le n \le 6$ and $h[n] = 1, for\ 0 \le n \le 4$ 8

   b) Define Z transform. Find the inverse z transform of $X(z) = \log(1 + az^{-1})$ 7

3. a) Why do we need DFT when we have DTFT? Determine the circular convolution of the sequence 8

$$x_1(n) = \{1, 2, 3, 4\}\ and\ x_2(n) = \{5, 6, 7, 8\}$$

   Also verify your answer.

   b) Define Fast Fourier Transform (FFT)? Determine 8-point DFT of the sequence using Decimation in Time FFT. 7

$$x(n) = cos\left(\frac{n\Pi}{4}\right)$$

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4. a) Obtain the parallel form realization of following IIR filter 7

$$H(z) = \frac{1 + \frac{1}{4}z^{-1}}{(1 + \frac{1}{2}z^{-1})(1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2})}$$

b) Convert the following IIR filter into lattice ladder structure 8

$$H(z) = \frac{1 + z^{-1} + 2z^{-2} + z^{-3}}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}}$$

5. a) Convert the following in Direct form- II realization described by 7

$H[z] = (1+z^{-1}+2z^{-2}+3z^{-3})/(1+15z^{-1}+15z^{-2}+12z^{-3})$

b) With the low pass specification $\Omega_p = 3.2$ KHz and $\Omega_s = 4.8$ KHz and sampling frequency $\Omega fs = 12$ KHz and $\alpha_s = 40$ dB, find the length and value of β for Kaiser Window. 8

6. a) Derive Impulse Invariant Method for IIR filter design. Also illustrate the mapping from the s-plane to the z-plane while using IIM. 7

b) Design a Chebyshev analog filter with maximum passband attenuation of 2.5 dB at $\Omega_p = 20\, rad/sec$ and stopband attenuation of 30 dB at $\Omega_s = 50\, rad/sec$ 8

7. Write short notes on: (Any two) 2×5

a) Gibbs Phenomena

b) Basis Elements of DSP

c) BIBO Stable

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POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the advantages of digital signal processing over analog signal processing? Briefly explain the basic elements of digital signal processing with the help of block diagram. 8

   b) Obtain a linear convolution of the following two discrete-time signals: 7
   $x(n) = \sum_{k=0}^{3} \delta(n-k)$ and $h(n) = 2^n[u(n)-u(n-3)]$

2. a) Define the term causality, stability, time-invariance and linearity of discrete time system. And state their significance in case of LTI system. 7

   b) Define Z transform. Find the inverse z transform of 8
   $X(z) = \log(1 + az^{-1})$.

3. a) Find the 8-point DIFFFT of $x(n) = sin\frac{\pi}{3}n$. 8

   b) Show that the multiplication of two DFT sequences results in circular convolution. 7

4. a) Find the direct form-I and direct form-II realizations of a discrete-time system represented by the transfer function 8
   $H(z) = \frac{3z^3-5z^2+9z-3}{[z-(\frac{1}{2})][z^2-z+(\frac{1}{3})]}$

   b) How are FIR digital filters designed using different approaches? How would you use a rectangular window to design a FIR filter? Explain. 7

5. a) Design a linear FIR filter using Kaiser window to meet the following specifications: 7
   $0.99 \leq |H(e^{j\omega})| \leq 1.01, for 0 \leq |\omega| \leq 0.19\pi$
   $\leq |H(e^{j\omega})| \leq 0.01, for 0.21\pi \leq |\omega| \leq \pi$

   b) Design a digital lowpass filter using Bilinear Transformation to satisfy the following requirements 8

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- Monotonic stopband and passband
- –3dB cut off frequency at 0.6π radians, and
- Magnitude down at 15 dB at 0.75π radians

6. a) Convert the analog filter with system function 8

$$H_a(S) = \frac{S + 0.2}{(S + 0.2)^2 + 9}$$

into a digital filter by means of the bilinear transformation. The digital filter is to have a resonant frequency of $\omega_r = \pi/2$.

b) Differentiate between FIR and IIR digital filters with examples. 7

7. Write short notes on: **(Any two)** 2×5

a) Linear convolution vs. circular convolution

b) Causal System

c) Remez Exchange Algorithm

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